Mean Fragility and Apparent Structure

A cautionary study, and a critical-thinking instrument, using Pascal, Delannoy, Motzkin, Catalan, and Worpitzky triangles

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Abstract

Graphs tend to speak before mathematics has had a chance to explain them, and a confident reader will often hear more than the data is entitled to say. This note works through a small empirical case in which ratios of arithmetic, root-mean-square, geometric, and harmonic means are formed from combinatorial triangle rows and diagonals, and in which the Pascal Fibonacci diagonals appear to display a striking period-two oscillation beneath a growing envelope. The picture is seductive because it borrows the visual grammar of phase oscillation, of group versus phase behaviour in waves, and even of Feigenbaum period doubling, yet a disciplined comparison across several means and across nearby combinatorial families shows that the effect is neither an averaging artifact nor a deep dynamical bifurcation. The substantive lesson is methodological, and it is the reason the piece exists as a teaching tool: an apparent graphical structure should be stress-tested across central means and across neighbouring families before anyone is permitted to infer a mechanism, and the same ladder of inference transfers far beyond combinatorics.

What this study is for

This document has two readers in mind at once, and it is worth being honest about the doubling. To one reader it is a short investigation in the combinatorics of triangle arrays, with a generating-function explanation at the end that settles the matter. To the other reader, the student working through the companion interactive, it is a controlled rehearsal of the moves that responsible quantitative reasoning actually requires, staged on an example small enough to hold in the hand but rich enough to be genuinely tempting. The mathematics is the vehicle; the transferable skill is learning to distrust a graph in a principled, rather than a merely sceptical, way.

The temptation at the centre of the study is one that physics students in particular will recognise, because the imagery is already in their heads. When a quantity oscillates between two values while its overall scale climbs, the trained eye reaches immediately for a wave packet, for the distinction between a fast phase that flickers and a slow group envelope that drifts, and a student who has met chaos will reach further still toward the period-doubling cascade that announces the onset of Feigenbaum dynamics. Each of these readings carries a whole theoretical apparatus behind it, and each would, if correct, license a great deal of further inference. The discipline this study teaches is the habit of asking, before any of that apparatus is invoked, whether the oscillation belongs to the object being measured or to the instrument doing the measuring.

The ladder of inference

The argument of this note climbs a fixed sequence of rungs, and the sequence is the thing worth memorising, because it is reusable wherever a graph appears to contain a pattern.

1. **Perception.** Name precisely what the graph seems to show, separating the envelope, the oscillation, and the late-time regime rather than letting them blur into a single impression.
2. **Temptation.** Name the theory the picture is inviting you to adopt, so that the invitation becomes visible and can be resisted on purpose.

3. **Instrument robustness.** Vary the measuring lens, here the choice of mean, and ask whether the pattern survives the change of instrument.
4. **Family robustness.** Vary the underlying object, here the combinatorial family, and ask whether the pattern is generic or specific.
5. **Generator.** Seek the mechanism, here a generating function and a recurrence, that would explain the pattern if it is real.
6. **Calibrated conclusion.** State the most that the evidence now permits, and no more, distinguishing what is real from what merely resembled something deeper.

The reason the ladder matters more than the particular answer is that its rungs are not specific to triangles. The same sequence governs a clinical trial that seems to show an effect, an economic series that seems to cycle, or a detector trace that seems to ring; in every case the responsible move is to vary the instrument, vary the population, and only then to propose a mechanism. A student who internalises the ladder on Pascal diagonals has acquired something that will outlast any particular fact about binomial coefficients.

1 The caution

A graph may show something real, and it may equally show something manufactured by the act of measurement, so the first task is to be clear about what the measuring instrument is. In this setting the instrument is not a physical detector but a choice of mean, and that choice is far from neutral. Given positive data $x_0, x_1, \ldots, x_{L-1}$, we compare four central means:

$$A = \frac{1}{L}\sum_i x_i, \qquad Q = \sqrt{\frac{1}{L}\sum_i x_i^2}, \qquad G = \left(\prod_i x_i\right)^{1/L}, \qquad H = \frac{L}{\sum_i x_i^{-1}}.$$

These quantities obey the classical chain

$$H \le G \le A \le Q,$$

yet the ordering conceals the more important fact that they do not see the same object. The arithmetic mean responds to additive mass, the root-mean-square mean leans toward the largest interior peaks, the geometric mean registers the multiplicative bulk of the whole profile, and the harmonic mean is acutely sensitive to the smallest boundary terms because it is built from reciprocals. Four lenses, four different things brought into focus.

Rule 1. If a pattern appears under only one mean, suspect the mean before you suspect the object.

Rule 2. If a pattern survives several means, do not yet relax; vary the underlying combinatorial family before claiming the pattern is intrinsic.

These two rules are the working form of rungs three and four of the ladder, and the rest of the study is essentially an extended demonstration that taking them seriously dissolves an apparent mystery into an ordinary, explicable effect.

2 The empirical observation

For each family we form a row or diagonal X_n, compute a mean M_n, and inspect the consecutive ratio

$$R_n = \frac{M_{n+1}}{M_n}.$$

The interactive that generates these ratios live, and that the companion worksheet walks through, is at

$$https://leelemacjames.github.io/pascal\_diagonal\_means\_learner.html.$$

and are in a pthyon workbook on google colab at: $googlecolabpythonnotebook$.

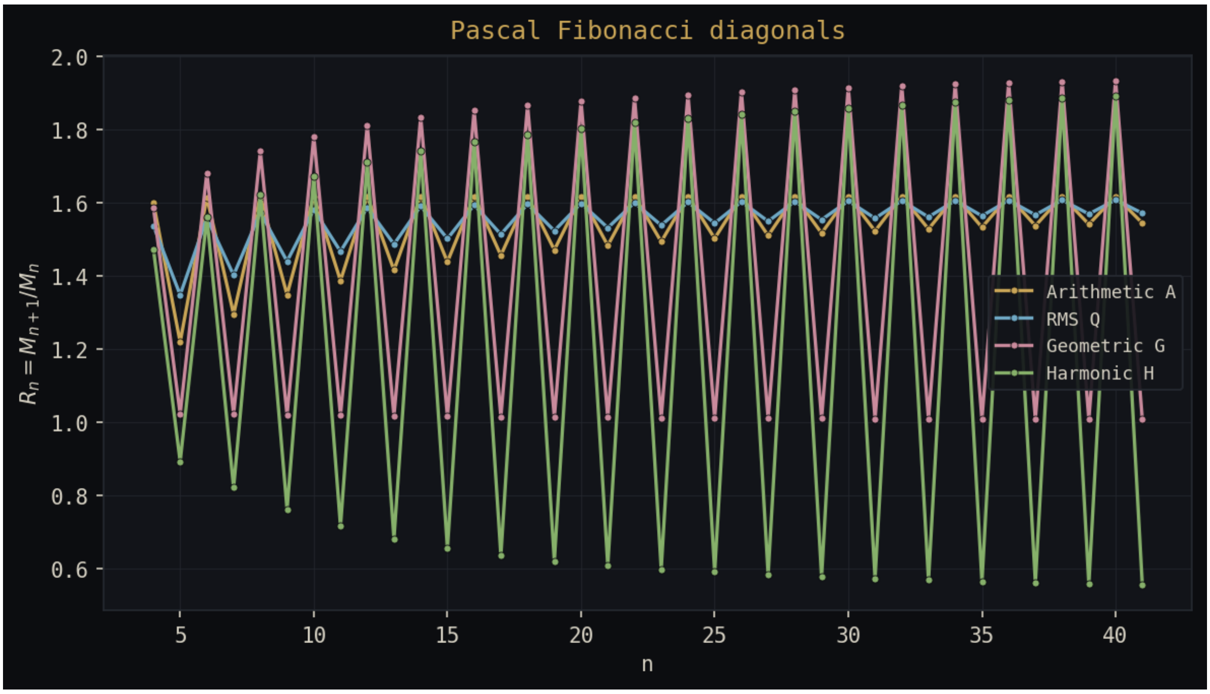


Figure 1: Graph of Pascal Fibonacci diagonals: ratios $R_n = M_{n+1}/M_n$ for arithmetic, RMS, geometric, and harmonic means suggesting an envelope, a phase oscillation, and a persistent two-cycle.

When the Pascal Fibonacci diagonals are plotted in this way, as in Figure 1, the eye is offered a great deal at once: a fast initial rise that reads as an envelope, a persistent oscillation that reads as a phase, a limiting band that reads as an attractor, a clean period-two alternation, and behind all of it a faint resemblance to the bifurcation diagrams associated with Feigenbaum period doubling. It is precisely because the picture offers so much that it is dangerous, and it is worth pausing on why the last analogy in particular must be refused. Feigenbaum period doubling is a statement about a family of dynamical systems indexed by a control parameter, in which successive bifurcations accumulate at a universal rate; here there is no control parameter being tuned, no dynamical system being iterated, and therefore nothing that the word "bifurcation" could honestly attach to. The visual similarity is real, and it is also, at this stage, mathematically empty. A safer and more honest description is simply that the Pascal ratios exhibit a stable parity split, a difference in behaviour between even and odd diagonals, whose status as signal or artifact has not yet been determined.

2.1 A small numerical window

The impression is already legible in a short window. Writing $R_n^{(M)} = M_{n+1}/M_n$ for the ratio under mean M, the Pascal Fibonacci diagonals give:

n	$R_n^{(A)}$	$R_n^{(Q)}$	$R_n^{(G)}$	$R_n^{(H)}$
20	1.618034	1.598788	1.877862	1.804017
21	1.483198	1.531560	1.012594	0.607943
22	1.618034	1.600460	1.887749	1.818565
23	1.493570	1.538375	1.011829	0.598166

The arithmetic and root-mean-square ratios sit close to a golden-ratio band with only a mild parity wobble, whereas the geometric and especially the harmonic ratios swing violently between even and odd n. This table is not a proof of anything, and it is important for the student to feel why: it is a warning that the graph is detecting something systematic, while different means are clearly responding to different aspects of the same underlying data, which is exactly the situation Rule 1 was written for.

2.2 Robustness across mean types

For the Pascal diagonals the arithmetic mean is unusually transparent, and tracing it carefully shows how much of the picture is already fixed before any subtle dynamics could enter. Because the shallow-diagonal sum of binomial coefficients is a Fibonacci number,

$$\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} = F_{n+1},$$

the arithmetic mean of the diagonal is

$$A_n = \frac{F_{n+1}}{\lfloor n/2 \rfloor + 1},$$

so that

$$\frac{A_{n+1}}{A_n} = \frac{F_{n+2}}{F_{n+1}} \cdot \frac{\lfloor n/2 \rfloor + 1}{\lfloor (n+1)/2 \rfloor + 1}.$$

The first factor tends to the golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$, and the second factor is a purely arithmetic parity correction arising from the floor function in the length of the diagonal. The arithmetic mean therefore already predicts the main shape, namely $R_n^{(A)} \approx \varphi \times (\text{parity correction})$, and it does so without invoking anything dynamical. The root-mean-square mean behaves similarly because it is dominated by the same large interior entries, the geometric mean shows a stronger two-cycle because it sees the entire multiplicative profile, and the harmonic mean is the most violently parity-sensitive of all because it is governed by the smallest boundary terms.

2.3 The exact geometry of the parity split

It pays to look directly at the diagonals, since the harmonic mean's behaviour then stops being mysterious. The relevant Pascal diagonals are

$$D_n = \left\{ \binom{n-k}{k} : 0 \le k \le \lfloor n/2 \rfloor \right\},$$

and separating them by parity gives, for the even case,

$$D_{2m} = \left(1,\ \binom{2m-1}{1},\ \binom{2m-2}{2},\ \ldots,\ \binom{m}{m}\right) = (1,\ 2m-1,\ \ldots,\ 1),$$

whose two boundary entries are both 1, while for the odd case,

$$D_{2m+1} = \left(1,\ \binom{2m}{1},\ \binom{2m-1}{2},\ \ldots,\ \binom{m+1}{m}\right) = (1,\ 2m,\ \ldots,\ m+1),$$

whose right boundary is $m+1$ rather than 1. The shorthand $D_{2m} = (1, \ldots, 1)$ should be read only as a claim about the two endpoints, not about the interior.

The harmonic mean is built from $H_n = L_n / \sum_i x_i^{-1}$, and the reciprocal sum is dominated by the smallest entries, which are exactly the boundary terms. In the even case the two boundary reciprocals contribute $1^{-1} + 1^{-1} = 2$, whereas in the odd case they contribute $1^{-1} + (m+1)^{-1} = 1 + \frac{1}{m+1} \to 1$. This single difference, between a boundary reciprocal mass of roughly two and roughly one, is the whole of the persistent even–odd pulse in the harmonic ratio. The oscillation that read so convincingly as a dynamical bifurcation is, on inspection, a boundary effect rendered visible by the choice to average reciprocals.

2.4 The generator

The mechanism, the fifth rung of the ladder, is a generating function. Defining the diagonal polynomial

$$D_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k} x^k,$$

one has the ordinary generating function

$$\sum_{n \geq 0} D_n(x)\, t^n = \frac{1}{1 - t - xt^2},$$

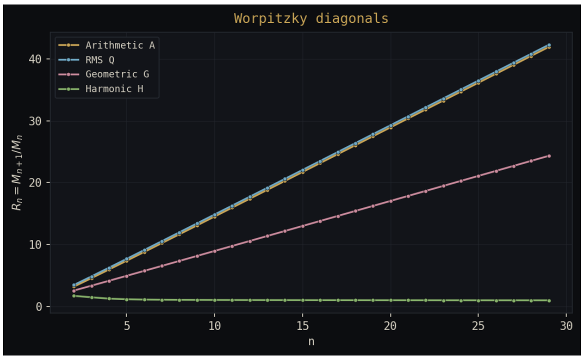
which at $x = 1$ returns $D_n(1) = F_{n+1}$ and so confirms that the additive mass of the diagonal is Fibonacci-governed. The same denominator $1 - t - xt^2$ encodes a second-order recurrence,

$$D_n(x) = D_{n-1}(x) + x\, D_{n-2}(x),$$

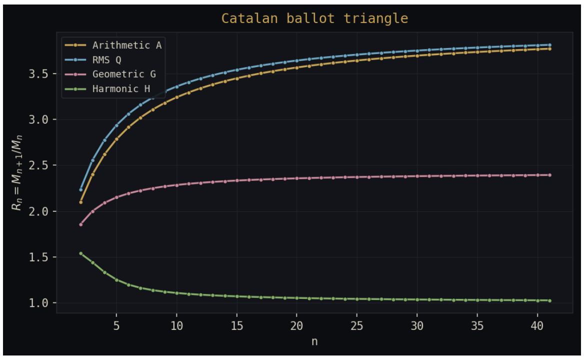
and it is this second-order, parity-aware structure that supplies the even–odd signature. The graph now has a complete account: Fibonacci growth fixes the envelope, the second-order recurrence supplies the parity structure, and the choice of mean determines how loudly that parity structure is heard.

3 Control families

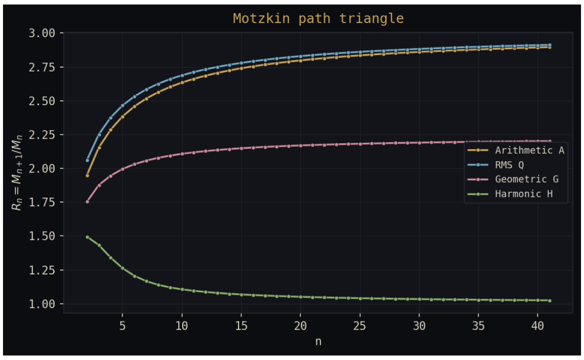
A graph can only ever be interpreted against alternatives, so the Pascal case is compared with four families chosen to sit at increasing distances from it. Catalan and Motzkin are close controls that retain path-counting or binomial ancestry, Delannoy is a broader lattice-path control, and Worpitzky is a deliberately distant control built from factorial and Stirling growth. The design of this comparison is itself part of the lesson, since a control is only informative to the degree that it shares the features one suspects of generating the pattern while differing in the feature one wishes to test.



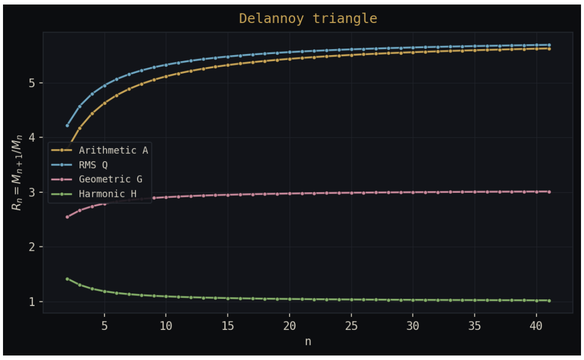
(a) Worpitzky diagonals



(b) Catalan ballot triangle



(c) Motzkin path triangle



(d) Delannoy triangle

Figure 2: Comparison of ratio dynamics for arithmetic, RMS, geometric, and harmonic means. Worpitzky, Catalan, Motzkin, and Delannoy all exhibit smooth convergence, and in particular none of them reproduces the persistent even–odd band seen in the Pascal case of Figure 1.

3.1 Catalan and ballot rows

The Catalan or ballot triangle,

$$C(n,k) = \binom{n+k}{k} - \binom{n+k}{k-1}, \qquad 0 \le k \le n,$$

inherits binomial structure under a non-crossing constraint, and so it is the right control for the question of whether Pascal ancestry alone manufactures the two-cycle.

3.2 Motzkin rows

Motzkin path rows obey

$$T(n,k) = T(n-1,k-1) + T(n-1,k) + T(n-1,k+1), \qquad T(0,0) = 1,\ T(n,k) = 0 \text{ for } k < 0,$$

and arise from paths with up, level, and down steps, testing whether a path family with a natural height boundary reproduces the parity split.

3.3 Delannoy rows

Delannoy rows obey

$$D(n,k) = D(n-1,k) + D(n,k-1) + D(n-1,k-1), \qquad D(n,0) = D(0,k) = 1,$$

with central growth constant $3 + 2\sqrt{2}$. Empirically the arithmetic and root-mean-square ratios approach this central-growth regime, the geometric mean approaches a lower multiplicative-bulk limit, and the harmonic mean approaches a boundary regime near unity, with the decisive observation being the absence of any persistent Pascal-like two-cycle.

3.4 Worpitzky rows

The Worpitzky rows,

$$W(n,k) = k!\, S(n+1, k+1),$$

carry factorial and Stirling growth, so their raw ratios grow roughly monotonically rather than settling to a small constant. As the most distant control, Worpitzky answers a narrow question, whether the Pascal oscillation is merely a consequence of taking means at all, and the answer is plainly no, while the closer Catalan, Motzkin, and Delannoy controls are needed to test the stronger claim that the phenomenon is specific to the Pascal Fibonacci diagonals. Across all four controls the parity oscillation simply fails to appear, which is what licenses the conclusion that it is a property of the Pascal generator rather than of averaging in general.

4 Interpretation

The experiment earns its keep by separating three things that the first graph had visually fused into one. There is an *envelope*, the early growth as a row or diagonal fills out; there is a *phase oscillation*, the even–odd alternation; and there is an *asymptotic regime*, the late limiting behaviour of R_n. The Pascal case happens to contain all three at once, which is exactly why it is so easy to misread, whereas Delannoy shows clean limiting behaviour with no persistent parity band, Worpitzky shows strong growth with no period-two structure, and Motzkin and Catalan provide the fairer intermediate tests that the close comparison demands.

This is the moment to be explicit about the distinction the study has been circling, between a genuine periodicity grounded in the structure of the object and a phase artifact that owes its visibility to the instrument. The even–odd split in the Pascal diagonals is real in the sense that it is dictated by the second-order recurrence and the parity of the diagonal length, so it is not an illusion; it is, however, not the dynamical period doubling its appearance suggested, because there is no parameter whose tuning produces a cascade. The honest reading is that a real, static parity structure has been amplified into something that looks dynamical by the reciprocal-sensitive harmonic mean, and recognising that difference is the whole point of the exercise.

5 A restrained conclusion

The original graph was neither meaningless nor sufficient on its own to support a foundational thesis, and the responsible chain of inference is the ladder set out at the start:

$$\text{graphical pattern} \longrightarrow \text{mean robustness} \longrightarrow \text{family robustness} \longrightarrow \text{generator explanation.}$$

For the Pascal Fibonacci diagonals the generator $1/(1 - t - xt^2)$ explains why Fibonacci growth appears, while the structural difference between the $2m$ and $2m+1$ diagonals explains why the ratios preserve an even–odd signature, and the comparison families establish that neither feature is a generic consequence of averaging. The warning that the whole study is built to deliver is that graphs are inclined to speak before they explain, and that a reader who answers their first sentence with a thesis has usually misheard them.

The teaching maxim

First see the structure, then try to destroy it. If it survives, it may deserve a theorem.

6 Using the learner and the worksheet

The companion interactive at `https://leelemacjames.github.io/pascal_diagonal_means_learner.html` is built so that each rung of the ladder corresponds to a control the student can move. The triangle-family selector is the family-robustness rung, the four mean toggles are the instrument-robustness rung, and the choice between the raw ratio R_n and the normalised ratio R_n/n lets the student strip out trivial growth so that any residual oscillation stands out cleanly. The recommended path through the tool is to begin with the Pascal family under the arithmetic mean, where the golden-ratio band and its mild parity wobble are easiest to read, then to add the geometric and harmonic means in turn and watch the parity split grow louder, and only afterwards to switch families and confirm that the loud parity behaviour vanishes everywhere except in Pascal. The accompanying worksheet is structured around exactly this path, and it asks at each stage not merely what the student sees but what inference the picture is tempting them to draw and what evidence would be needed to earn it. The intended outcome is a student who, faced with the next seductive graph in whatever subject, reaches first for the instrument and the control rather than for the theorem.